



National Teachers College
629 J Nepomuceno, Quiapo, Manila, 1001 Metro Manila
Bachelor of Science in Information Technology

Final Project/Exam: System Analysis and Design (SAD) for Sustainable Development

System Analysis and Design

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BSIT - 2.3

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Software Requirements Specification (SRS)

Project Charter

Let's face it; the current process of scholarship application is simply chaotic. Scholarship applications made on paper tend to be lost. The registrar's office gets buried in mountains of paper each semester. There is no record for administrators to determine whose work has been reviewed and who is responsible for it. Sponsorship organizations do not know how the money allocated is used. This motivated us to create GrantEase.

Our main goals are simple:

What We Want	Why It Matters
Go completely digital	No more lost documents (we're losing about 15% right now)
Let students track their application	So they stop calling the registrar's office every day asking "is it approved yet?"
Keep student data safe and private	Because RA 10173 is not a suggestion – it's the law
Track every single data access	So we always know who looked at what, and when

How This Helps the World (SDG 4)



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This may seem ambitious, but in reality, our tiny project actually contributes to a UN objective, namely Quality Education for All (SDG 4).

The rationale is simple: nowadays, children from poorer households usually fail to apply for scholarships due to the overly bureaucratic procedure involved or simply not knowing whether their applications have reached the organization. With all the information stored electronically, however, such issues will no longer arise.

SMART Success Criteria

We wanted to be realistic about what "success" actually looks like. So we set specific targets:

Goal	Our Target	How We'll Know
Faster processing	From 14 days to 3 days	We'll measure the time between submission and decision
System reliability	99.9% uptime during deadlines	Our server logs will tell us
User training	100% of staff trained within 2 weeks	Training sign-off sheets
Zero lost applications	From 15% loss to 0%	Semester-end audit



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Working prototype	By June 30, 2026	Deployment sign-off from the school
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High-level Risks

We're not naive. We know things might not go perfectly. Here's what we're worried about and how we plan to handle it:

Risk	How Worried?	Our Backup Plan
Some students aren't tech-savvy or don't have scanners	Pretty worried	We'll put scanning kiosks on campus and accept phone camera photos
We accidentally violate data privacy laws	A little worried (but this is serious)	Have a lawyer review our work at every stage; encrypt everything
The server crashes right when deadlines hit	Moderately worried	Use auto-scaling cloud servers; test before deadlines
Registrar staff don't want to use our system	Somewhat worried	Hold workshops; let them test it first and give feedback



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Stakeholder Analysis

Interest/Influence Matrix

Who Are They?	What Do They Want?	How Much Do They Care?	How Much Power Do They Have?
Students	To know their application status; fair treatment; easy-to-use system	High	Low (they just want things to work)
Registrar Staff	Accurate grade verification; not too much extra work	High	High (they can make or break adoption)
Scholarship Admin	Transparent selection process; audit trail for every decision	High	High (they approve the funding)



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Sponsors/Donors	Proof that their money is being used properly	High	Medium (they provide the funds)
IT Support	Stable system; easy backups	Medium	Medium (they maintain everything)

Negotiation Log

What Was the Conflict?	Who Was Fighting?	How Did We Solve It?	When?
Students want real-time updates, but IT worries about server load	Students vs. IT Support	We compromised: every 5 minutes during busy times, instant during quiet times	Week 2
Registrar wants high-quality scans, but students only have phone cameras	Registrar vs. Students	We accept phone photos but flag low-quality ones for follow-up	Week 3



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Who does what? Registrar vs. Scholarship Admin role confusion	Both admin types	We made them completely separate in the system design – different logins, different dashboards	Week 4
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Requirements Elicitation

The Questions We Asked

We sat down with actual users and asked them these questions:

#	Question	Who We Asked
1	"What's the most frustrating thing about applying for scholarships right now?"	Students, Registrar
2	"What information do you wish you could see right now but can't?"	Scholarship Admin
3	"If you could change ONE thing about how applications are processed, what would it be?"	Registrar



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4	"How do you currently track which students got scholarships?"	Scholarship Admin
5	"What would make you trust a digital system with student data?"	Everyone
6	"How much delay in status updates is okay with you?"	Students
7	"What reports would you need to show sponsors?"	Scholarship Admin

Raw Notes



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Who Said It	Exact Words
Student A	"I submitted my application last month. I still don't know if they even got it."
Student B	"The deadline is 5 PM but the registrar's office closes at 4. I literally had to run."
Registrar Staff 1	"We get over 200 applications every week. The pile of paper on my desk never disappears."
Registrar Staff 2	"Sometimes students fake their grades. We need a way to check directly."
Scholarship Admin	"I need to see WHO reviewed WHICH application and WHEN. Right now, there's zero accountability."
IT Support	"Whatever we build has to work on the school's existing Wi-Fi. No budget for new infrastructure."



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Specific Requirements

Functional Requirements (The "System Shall" Statements)

We prioritized everything using MoSCoW (Must, Should, Could, Won't):

ID	What the System Must Do	Priority	Where This Came From
FR-01	Let students submit applications and upload documents digitally	MUST	Student A's complaint
FR-02	Send a confirmation receipt within 60 seconds of submission	MUST	Every student ever
FR-03	Show real-time status (Pending, Under Review, Approved, Rejected)	SHOULD	Student B's anxiety
FR-04	Let Registrar mark applications as Compliant or Non-Compliant	MUST	Registrar Staff 1
FR-05	Let Scholarship Admin review and approve/reject applications	MUST	Scholarship Admin



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FR-06	Keep an unchangeable log of every access, change, or decision	MUST	Scholarship Admin (accountability!)
FR-07	Generate anonymized funding reports for sponsors	COULD	Q7 from our interview
FR-08	Enforce role-based access (Registrar, Scholarship Admin, Sponsor are separate)	MUST	IT Support, Legal requirements
FR-09	Encrypt all personal information using 256-bit encryption	MUST	RA 10173 (Data Privacy Act)
FR-10	Let IT export audit logs for compliance checks	SHOULD	Legal Officer

Non-Functional Requirements (The Quality Stuff)

ID	Quality Requirement	Priority	How We'll Measure It
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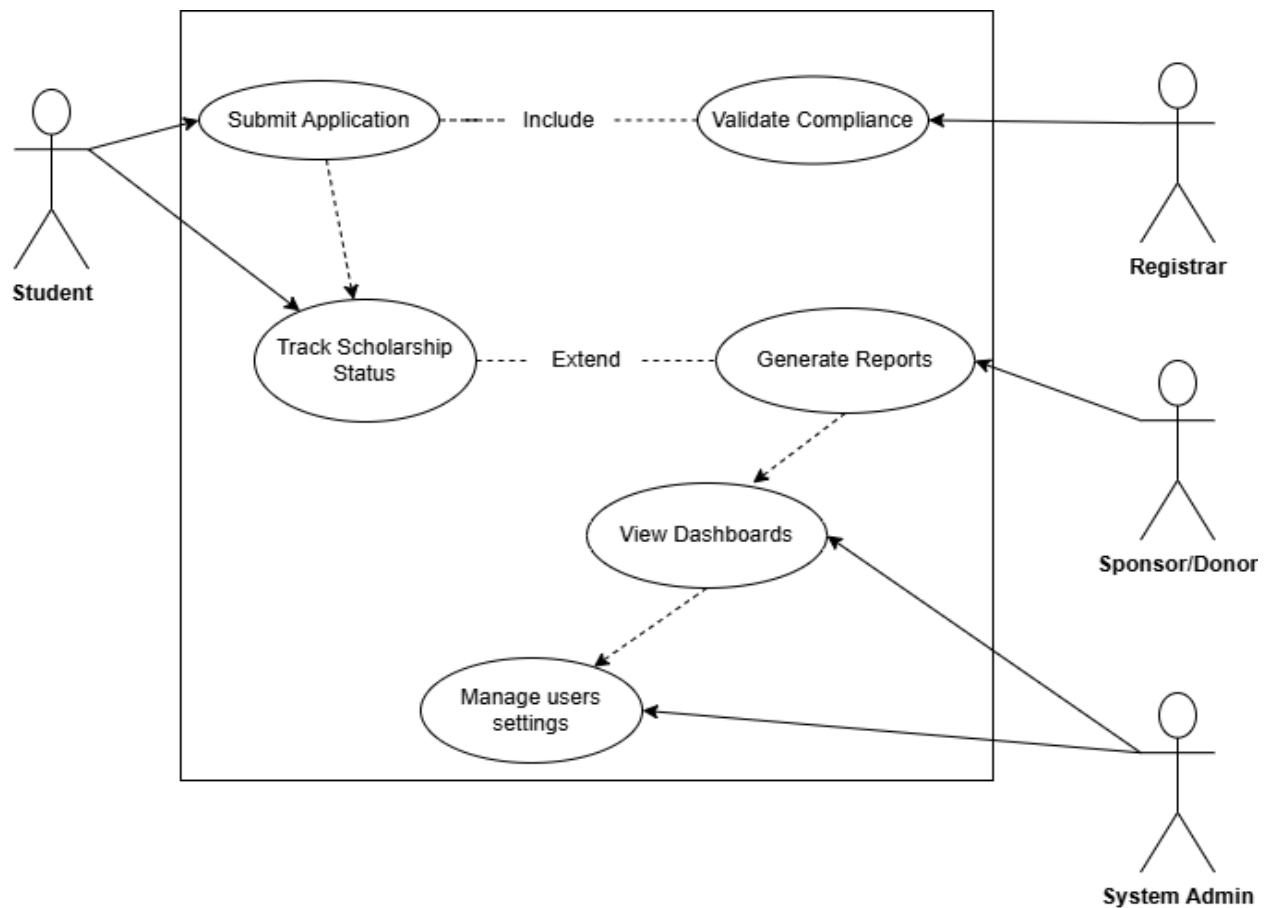
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NFR-01	System stays up 99.9% of the time during deadline periods	MUST	Monthly uptime report
NFR-02	Any page loads within 3 seconds on school Wi-Fi (5 Mbps)	SHOULD	Load testing
NFR-03	Encrypt data at rest with 256-bit AES, in transit with TLS 1.3	MUST	Security audit
NFR-04	Handle 500+ concurrent users during peak submission	MUST	Load test to 1,000 users
NFR-05	Keep audit logs for 5 years (legal requirement)	MUST	Database retention policy
NFR-06	Work on Chrome, Firefox, Edge, mobile and desktop	SHOULD	Cross-browser testing

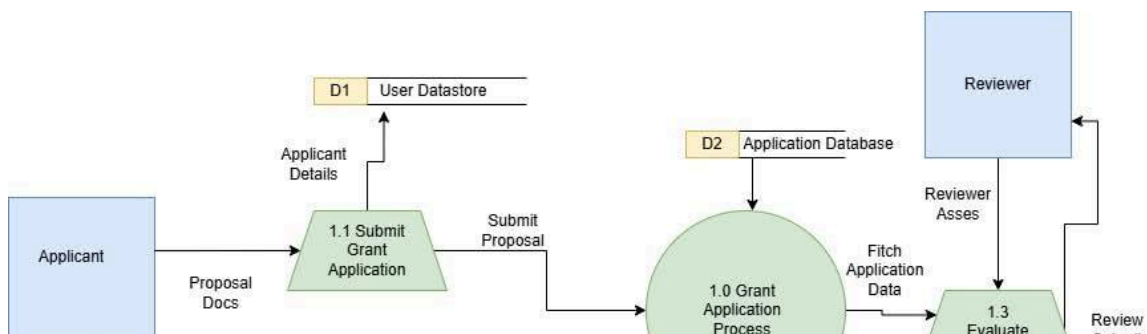


NFR-07	Recover from database failure within 15 minutes	COULD	Disaster recovery drill
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Logical Use Case Models



System Boundary





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